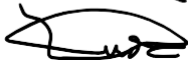


Royal Weave Report



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1. Introduction

This report presents the full development journey of Royal Weave, a shopping web-based system that started as a front-end prototype using HTML and CSS, then expanded into a more functional e-commerce website using PHP, JavaScript and a database. The project was designed to simulate the main features of an online shopping platform, including product browsing, product details, cart management, checkout, user accounts, purchase history and an admin control panel.

2. Overview

Royal Weave provides a simple and intuitive shopping experience where users can browse products, view product details, manage their cart and complete checkout. The system also includes user account features such as login, signup, profile editing and purchase history. On the admin side, the website includes a control panel that allows admins to manage products, update product information, control product status and search for products efficiently.

3. Database

3.1 Database design overview

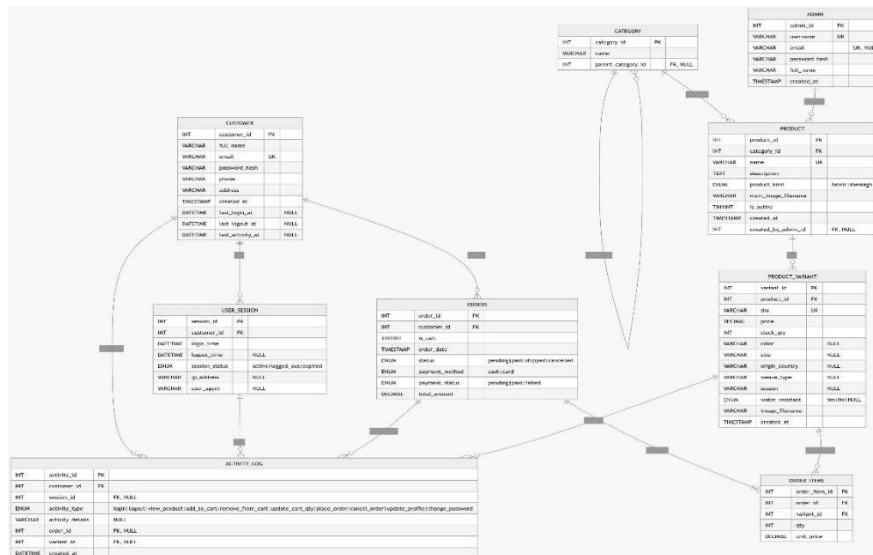
The database is designed using a relational model to ensure data consistency and integrity. It supports all core functionalities of a database system. Managing users, admins, products and orders. Design follows a normalized principle reducing redundancy and aims to reach an efficient relationship.

3.2 Tables Structure

Databases consist of interconnected tables with each serving a role. In royal weave we designed the table structure to separate responsibilities across user, product and order management.

- **Admin and Customer tables:** These tables store account information for both admins and users. Crucial for separating system control.
- **Category and Product tables:** These tables organize product data and classify them into categories.
- **Product Variant Table:** This table stores variant details such as price, stock quantity, size, color and image.
- **Orders and Order items tables:** These tables store the checkout process by storing cart information, completed orders, quantities and prices.

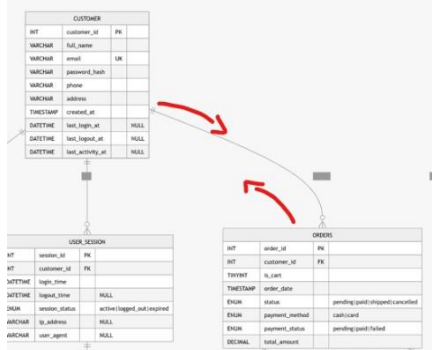
- **User session and Activity Log tables:** These tables record and store login sessions and activities. Such as viewing products, adding items to carts and placing orders. Tracking user behavior ensuring better experience.



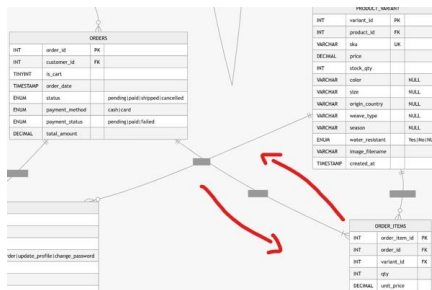
3.3 Entity-Relationship Design

The Entity-Relationship design explains how tables interconnect, using primary and foreign keys. Relationships are Crucial to ensure that the database runs as one mind.

- **Customer to Orders relationship:** One customer can create multiple orders.



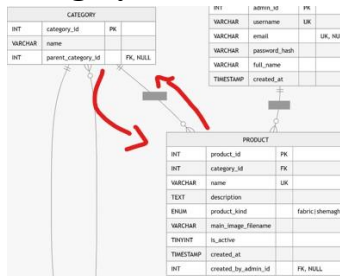
- **Orders to Order items relationship:** One order can contain many items.



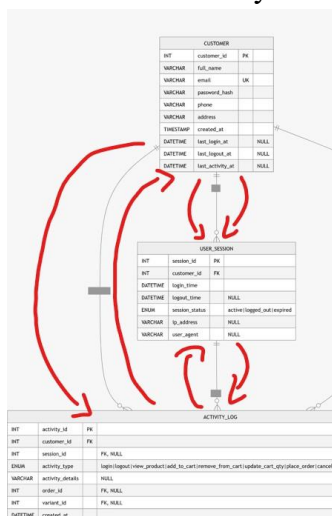
- **Product to Product Variant relationship:** One product can have multiple variants.



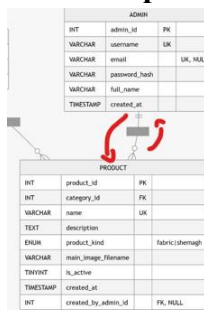
- **Category to Product relationship:** Each product belongs to a category.



- **Customer to Session and Activity Log relationships:** A customer can have multiple sessions and activity records.



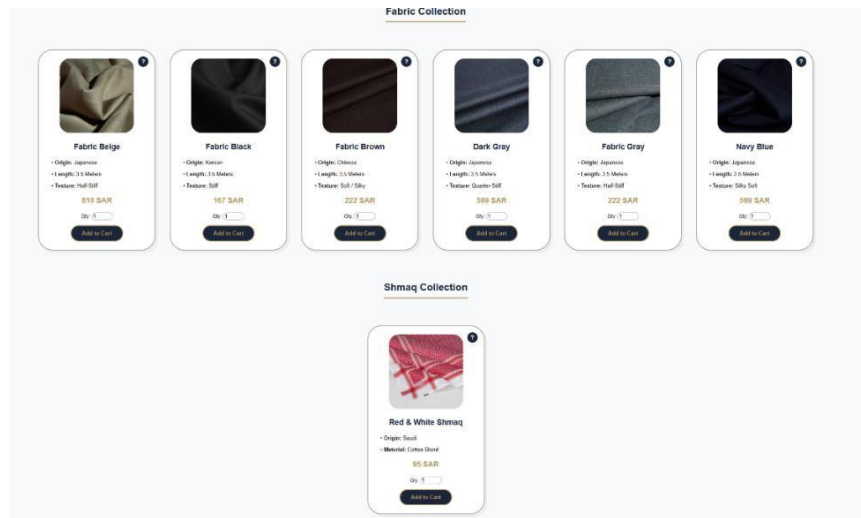
- **Admin to product relationship:** An Admin can create or manage products.



4. User Features

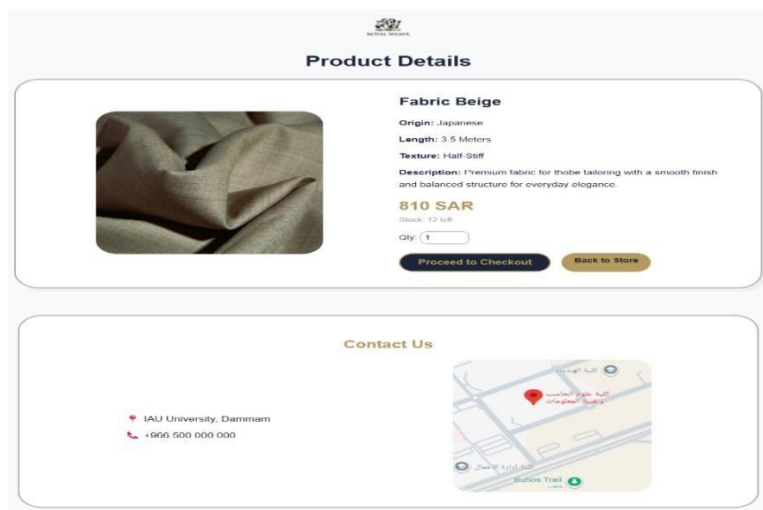
4.1 Product Listing

User has a display of varieties of available products in an organized fashion, where each product includes an image and basic information. A cart icon is also displayed allowing fast access to check out. Ensuring ease and fast browsing.



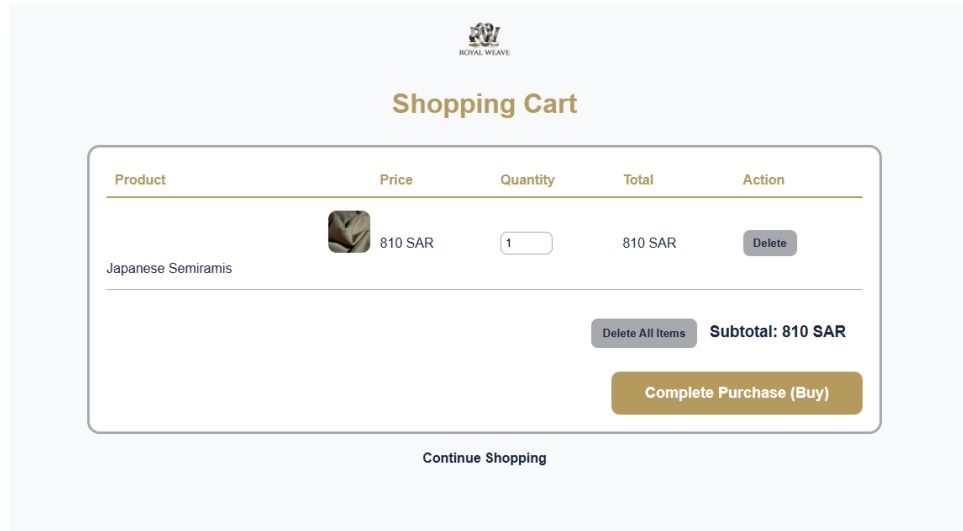
4.2 Product Details Page

When the user clicks on a product, they are directed to a detailed view of the selected product providing comprehensive information. Includes price, availability, and help.



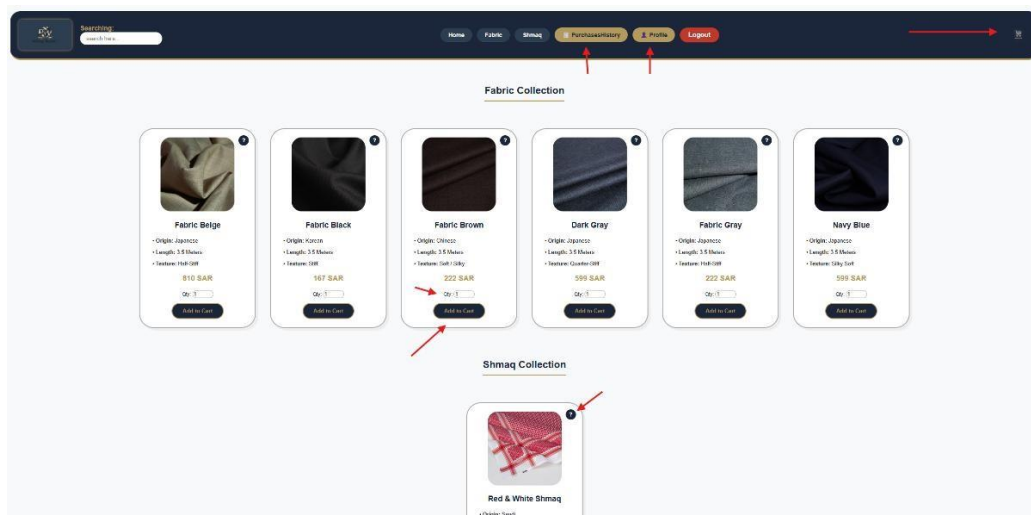
4.3 Cart and Checkout

The cart and checkout pages allow users to manage selected items in one place. Users can modify item quantities, delete individual items, clear the cart or proceed with selected items. The cart is connected to the database using PHP, allowing products added by the user to be stored and retrieved dynamically. JavaScript also supports the cart interface and improves user interaction.



4.4 Navigation Flow

The navigation flow is designed to be as straightforward as possible. Ensuring a user-friendly environment. Clicking on a product directs the user to a details page, from which they can proceed to check out. Also, users can access the checkout page and add the selected product to the cart by simply clicking on the cart icon. Additionally, a Contact Us page is accessible through the menu.

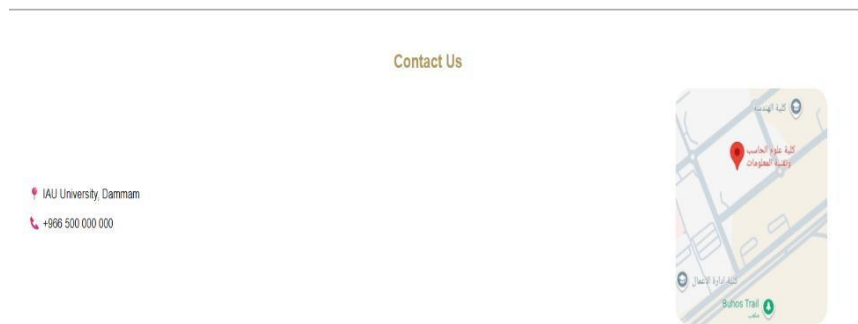


4.5 User Account and Purchase History

The user side includes login, signup, profile editing and purchase history pages. These features allow users to access their accounts, update their personal information and view previous purchases. This makes the website more complete and closer to a real e-commerce system.

4.6 Contact Us Page

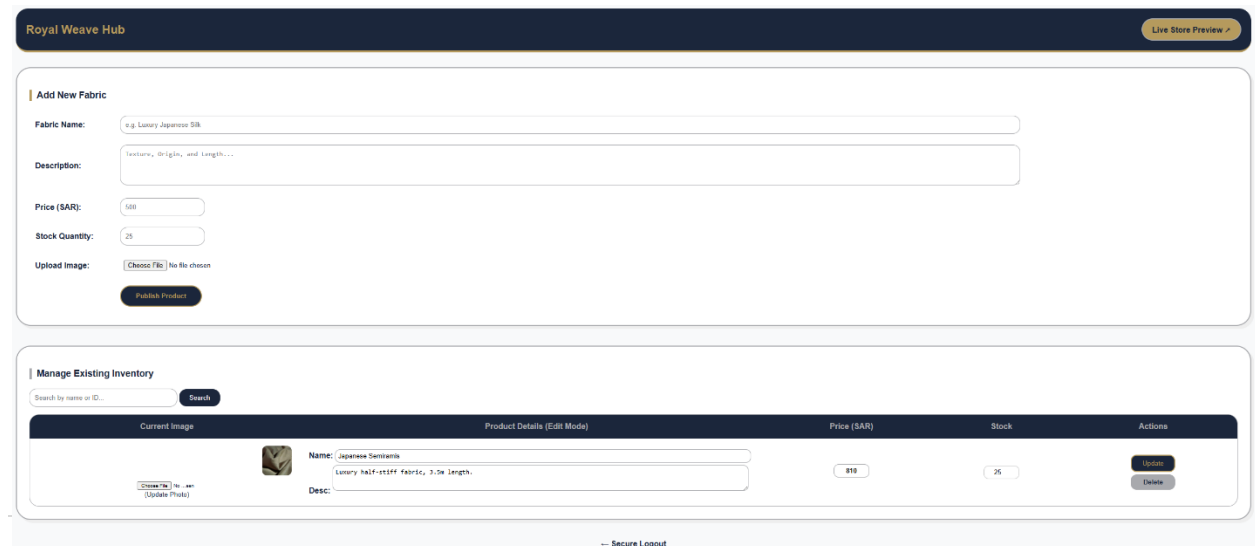
Contact Us page gives the user information regarding the location and a map for better visualization. Enhancing credibility.



5. Admin Features

5.1 Product Management

Admin control panel has full control over product management. An Admin can add new products, add their details and upload the images needed. Also, modifying existing products that includes updating their images and deleting unneeded products.



5.2 Search Functionality

Admin control panel includes a special search feature that allows an admin to quickly locate specific products. Improving efficiency when managing large numbers of products.

5.3 Admin Navigation Flow

The admin navigation flow begins with accessing the admin control panel, then logging in using the admin credentials. From there, the admin can perform actions such as adding, modifying or deleting products.

The screenshot displays the 'Royal Weave Hub' admin interface. At the top, there's a dark blue header with the site name and a 'Live Store Preview' button. Below the header, the 'Add New Fabric' section contains a form with fields for 'Fabric Name' (with a placeholder 'e.g. Luxury Japanese Silk'), 'Description' (placeholder 'Texture, origin, and length...'), 'Price (SAR)' (set to 500), and 'Stock Quantity' (set to 25). There's an 'Upload Image' section with a 'Choose File' button and a 'No file chosen' status. A 'Publish Product' button is at the bottom of this form. Below this, the 'Manage Existing Inventory' section features a search bar with a 'Search' button. A table lists existing products. The first row shows a product with a 'Current Image' placeholder, a 'Product Details (Edit Mode)' section with 'Name: Japanese Serikata' and 'Desc: Luxury half-silky fabric, 3.5m length...', 'Price (SAR)' of 810, and 'Stock' of 25. The 'Actions' column for this row has 'Update' and 'Delete' buttons. At the bottom, there's a 'Secure Logout' link.

Current Image	Product Details (Edit Mode)	Price (SAR)	Stock	Actions
 Click here to upload (Update Photo)	Name: Japanese Serikata Desc: Luxury half-silky fabric, 3.5m length...	810	25	Update Delete

5.4 Admin Input Validation

JavaScript validation was added to the admin page to prevent incorrect product data from being submitted. The validation checks that required fields are not left empty, product names do not contain only numbers, category selection is valid, prices and quantities are positive values and invalid negative numbers such as -1 are rejected. This helps reduce incorrect input before the form reaches the PHP and database side.

5.5 Admin Logout

After completing the necessary tasks, the admin can securely log out of the system. Maintaining security and preventing unauthorized access.

6. Fulfilled Requirements

6.1 Description

It successfully implements the core functionalities of an e-commerce web application. Users can browse products, view detailed product information, add products to the cart and complete the checkout process. The system also supports user account functionalities including login, signup, profile editing and purchase history viewing.

On the admin side, the control panel allows administrators to add, edit, delete and manage products dynamically using PHP and the database system. Product search functionality was also implemented to improve product management efficiency.

6.2 JavaScript Validation

JavaScript validation was integrated into the admin page to ensure that invalid data cannot be submitted easily. The validation checks include preventing empty fields, invalid category selection, number-only product names and negative values such as “-1” in price or stock quantity fields.

6.3 Tasks That Are Not Included

Task-1: Online payment gateway integration

Task-2: Email verification system

Task-3: Product recommendation system

Admin validation message

Product management page

Login field

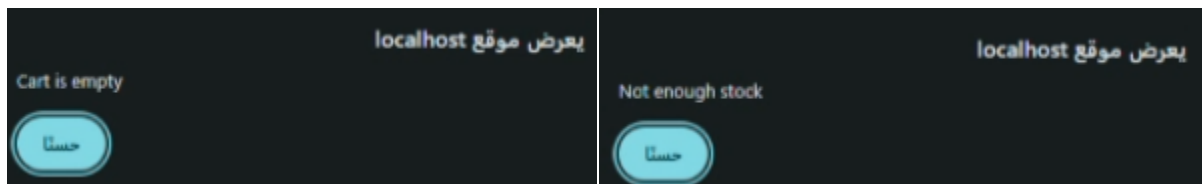
7. Testing Cases

The system was tested using multiple scenarios to ensure that the main functionalities operate correctly. Testing focused on both user-side and admin-side features.

7.1 Testing Table

Feature	Test Case	Expected Result	Actual Result
User Login	Enter correct credentials	User logs in successfully	Passed
User Login	Enter incorrect password	Error message displayed	Passed
Product Listing	Open products page	Products display correctly	Passed
Product Details	Click product	Product details page opens	Passed
Cart System	Add product to cart	Product added successfully	Passed
Cart System	Remove product from cart	Product removed successfully	Passed
Checkout	Confirm order	Order stored successfully	Passed
Admin Product Add	Valid product input	Product added successfully	Passed
Admin Validation	Empty required fields	Submission prevented	Passed
Admin Validation	Negative quantity	Validation error displayed	Passed
Search Functionality	Search for product name	Matching products displayed	Passed

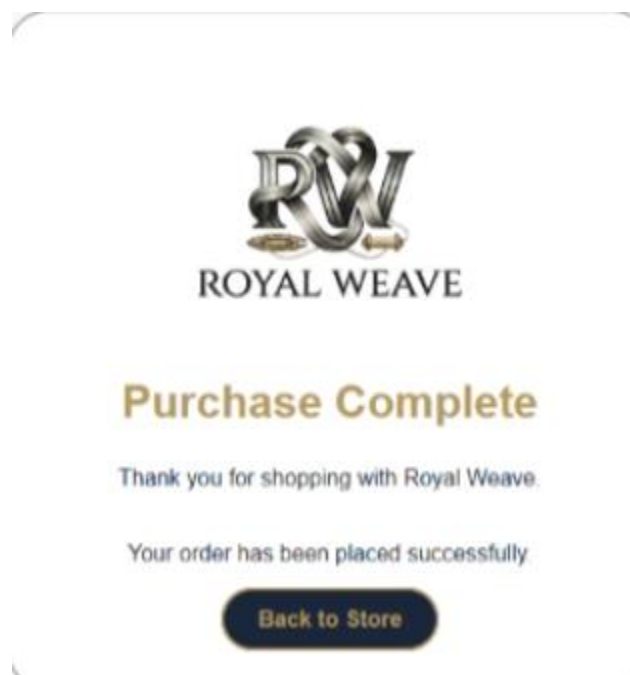
The testing process confirmed that the core project requirements function correctly and that validation mechanisms reduce incorrect admin input.



Validation

Product	Price	Quantity	Subtotal	Action
Group 5	 2.00 SAR	1	2 SAR	Delete
				Delete All Items Total: 2 SAR
				Complete Purchase (Buy)

Cart page



Successful checkout

Product Details



شششش

Origin:

Length: 3.5 Meters

Texture:

Description: شششش

111.00 SAR

Stock: 9 left

Qty:

Add to cart

Back to Store

Details

Searching:

Group 5

Fabric

Shmaq

Profile

PurchasesHistory

Fabric Collection

Shemagh Collection



Group 5

• Origin: ش

• Material: شش

2.00 SAR

Stock: 2 left

Qty:

Add to Cart

Search Function

Product	Price	Quantity	Subtotal	Action
Group 5 	2.00 SAR	1	2 SAR	<button>Delete</button>
				Delete All Items Total: 2 SAR Complete Purchase (Buy)

Cart

Fabric Collection



- Origin: البحرين
- Length: 3.5 Meters
- Texture:

250.00 SAR

Stock: 50 left

Qty:

Add to Cart



البحرين

- Origin:
- Length: 3.5 Meters
- Texture:

111.00 SAR

Stock: 0 left

Qty:

Add to Cart



abdulmohsen khalid

- Origin: البحرين
- Length: 3.5 Meters
- Texture: ممشق

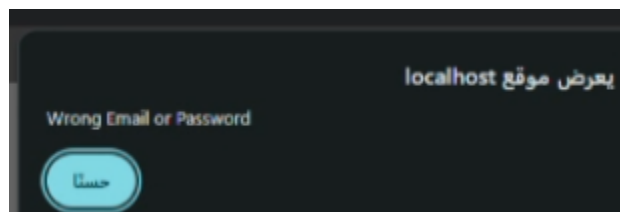
5.00 SAR

Stock: 4 left

Qty:

Add to Cart

Collection



Email Validation

Add New Fabric

Product Name:

Description:

Price (SAR):

Stock Quantity:

Product Type:

Origin Country:

Texture / Material:

Upload Image:

Product Preview

Product name must be 3-30 characters and contain letters.

Admin Validation

Enter Your Email:
admin1@gmail.com

Enter Your Password:
[password field]

☐ Remember Me

Log In

Continue as Visitor

⚠️ يرجى ملء هذا الحقل.

8. Project Implications

8.1 Description

It provides practical experience in developing a complete web-based shopping system by integrating front-end design, database systems, PHP and JavaScript functionality. The project helped improve understanding of how dynamic web applications operate and how different technologies communicate together.

8.2 Development process

The development process also strengthened teamwork and collaboration skills, as team members worked on different parts of the project including front-end development, database design and interface functionality.

8.3 Future Expansion

It can be expanded in the future into a more advanced e-commerce platform by integrating secure payment systems, improved security mechanisms and additional customer features.

9. Teamwork and Contribution

We as group 7 worked together in a structured yet flexible manner, ensuring all parts of the project were well executed and integrated. Turki and Abdulmohsen focused primarily on developing the HTML structure, with Turki contributing to the report writing. Ziyad and Tamim were responsible for designing the database. Mohammed handled the CSS styling, creating a consisted and user-friendly interface. Despite these primary roles, we as a team maintained a strong collaboration on the project, with all members contributing and reviewing each other's work. During the later implementation stage, the project was expanded with PHP, database connection and JavaScript functionality. The JavaScript contribution focused especially on improving the admin page through input validation, search support and better interaction, while the PHP and database parts handled dynamic system operations. Despite these primary roles, the team maintained strong collaboration, with all members contributing and reviewing each other's work.

10. Conclusion

Royal Weave demonstrates the core features of a shopping web-based system. The project started as a front-end prototype and developed into a more complete e-commerce system with dynamic product management, user account features, cart and checkout functions, purchase history and admin-side validation.